INCH-POUND

MIL-S-83807 (84)
27 Apr 90

MILITARY SPECIFICATION

SCRAPER, SELF PROPELLED, 11 CUBIC YARDS CAPACITY, SELF ELEVATING

This specification is approved for use within WR-ALC, Code 84
Department of the Air Force, and is available
for use by all Departments and Agencies of the
Department of Defense.

1. SCOPE.

1.1 Item. This specification covers a commercial, self propelled, self elevating earth moving scraper of at least 11 cubic yards heaped capacity.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Military

MIL-T-83133	Turbine Fuels, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

Federal

FED-STD-595	Colors.

Military

MIL-STD-130	Identification Marking of US Military Equipment.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1474	Noise Limits for Army Material.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3805

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

USAF DRAWING 8838302	Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicle and Support Equipment.

(Drawing can be obtained from WR-ALC/MMDEC, Robins AFB, GA. 31098-5609)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

The Tire and Rim Association Inc Yearbook

(Application for copies should be addressed to The Tire and Rim Association Inc, 3200 West Market St, Akron OH 44313.)

Society of Automotive Engineers

SAE J276	Steering Frame Lock for Articulated Loaders and Tractors.
SAE J298	Universal Symbols for Operator Controls on Industrial Equipment.
SAE J534	Lubrication Fittings.
SAE J737	Hole Spacing for Scraper and Dozer Edges.
SAE J738	Scraper and Dozer Edge Cross Sections, Rolled, Cast, Forged, or Machined.
SAE J957	Elevating Scraper, Capacity Rating.
SAE J1040	Performance Requirements for Rollover Protective Structures for Construction, Earthmoving, Forestry, and Mining Machines.
SAE J1152	Breaking Performance, Rubber Tired Construction Machines.

(Application for copies should be addressed to The Society of Automotive Engineers, 400 Commonwealth Dr, Warrendale PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Description. The scraper shall be the conventional two axle, pusher plate, articulated steer, front axle drive, diesel engine driven, self-elevating type and shall have a minimum heaped capacity of 11 cubic yards. The unit shall be designed for a payload not less than 26,000 pounds.

3.1.1 Commerciality. The scraper shall be the latest model of the standard product of the contractor and shall have demonstrated industry acceptability by having been manufactured, marketed, and sold in significant numbers to the commercial public. Product improvements incorporated in the commercial production unit during this one year period are acceptable. All items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

as standard equipment, shall be furnished. The scraper shall be equipped with optional equipment as specified herein. Optional equipment is defined as equipment not standard with the scraper but which has been furnished to the commercial customer in the past, such as special features and allied equipment.

3.1.2 Completeness. All units shall be delivered completely assembled, adjusted, and serviced (except for fuel). If required for transport, the units shall be reduced to transport configuration.

3.1.3 Operating conditions. Unless otherwise specified, the scrapers must be capable of extended storage and operation in all of the following environments and conditions.

a. Temperatures-	Engine starting	0°F -	110°F
	Unrestricted operation	-20°F -	110°F
	Storage	-40°F -	125°F

If additional winterization is specified (see 6.2), cold temperatures shall be as specified in 6.2. Cold temperatures are that of ambient air without the benefit of solar radiation. Hot temperatures are that of ambient air. Hot temperature performance shall include the impact of full tropical solar radiation.

b. Scraper shall not be damaged or any function impaired by wind blown sand, dust, or precipitation.

c. Scraper shall withstand prolonged exposure to corrosive coastal environments.

3.1.4 Life. The scraper shall be designed and constructed to provide a minimum of twelve years service. The ability to cost effectively maintain a safe and efficient vehicle constitutes a serviceable vehicle.

3.2 Maintainability. All maintenance described in this specification shall be capable of being performed by one mechanic using common hand tools. Design and construction of the scraper shall permit ready accessibility to all parts for maintenance and repair in the field. Replacement and adjustment of component parts and accessories shall be accomplished with minimum drainage of fluids and disturbance to other parts.

3.2.1 Filling, draining, and checking provisions. All fluid systems shall possess a convenient means of filling, draining, and checking the level of the system. Fluid level indicators or fill ports shall identify the type of fluid, state whether the level should be checked cold, hot, or with engine running, as applicable, and identify appropriate add and full levels. Engine oil and transmission fluid reservoirs shall have a self-sealing sampling valve which produces a true sample of the circulating fluid.

3.2.2 Lubrication points. Lubrication points shall be provided for all surfaces requiring lubrication. Pressure relief fittings shall be supplied where the use of pressure lubrication equipment may damage seals. Lubrication fittings shall be in accordance with SAE J534 and be located so as to be clearly visible and easily serviced with a grease gun with a 10 inch flexible hose.

3.2.3 Positioning. All parts which could cause improper performance if positioned incorrectly shall possess asymmetrical mounting devices which prevent incorrect positioning.

3.2.4 Starter disconnect. The ignition system shall possess a starter disconnecting system which shall protect the cranking system by preventing the cranking system from engaging while the engine is running or rotating.

3.2.5 Down shift inhibitor. The transmission shall possess a down shift inhibitor to prevent damage and over revving of the engine and drive train by preventing unsafe down shifts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

3.2.6 Blade edge. Scraper blade edge shall be of 3 or 4 sections. All sections of the same length shall be interchangeable. All sections shall be reversible. Blade mounting shall be in accordance with SAE J737 and SAE J738.

3.2.7 Test points. Steel fittings shall be provided in the hydraulic system for attachment of a hydraulic system pressure tester. One test point fitting shall be installed upstream and adjacent to the filter, and one fitting shall be connected to the outlet of each pump.

3.2.8 Maintenance operations list. Any four or more operations shall be able to be performed by one person in the sum of times allotted for each operation.

a. Remove, replace, and adjust all engine driven belts - 1 hour.
b. Remove and replace voltage or current regulator - 15 min.
c. Remove and replace alternator and adjust alternator belts - 30 min.
d. Remove and replace all filter elements, screens, and strainers on the hydraulic system - 30 min.
e. Drain cooling system, remove and replace engine cooling hoses, and refill cooling system - 1 hour.
f. Drain engine lubricating oil, remove and replace oil filter elements, and refill crankcase - 30 min.
g. Remove and replace fuel filter elements - 30 min.
h. Remove and replace transmission fluid filter elements and strainer - 45 min.
i. Drain and refill transmission and torque converter - 20 min.
j. Drain and refill final drives - 30 min.
k. Disconnect battery leads, remove and replace batteries, and reconnect battery leads - 15 min.
l. Adjust elevator chain tension - 30 min.

3.2.9 First article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with section 4.4.

3.3 Reliability.

3.4 Safety. All exposed parts, subject to high operational temperatures or electrically energized, shall be located, insulated, enclosed, or guarded so as to prevent hazards to operating personnel. All moving parts which present a hazard shall be guarded.

3.4.1 Steps, grabrails, and platforms. Steps, grabrails, platforms, etc, shall be provided to ensure safe and convenient access to all control, maintenance, and inspection points. Steps, footholds, and walkways shall be expanded metal, perforated metal, or possess a durable nonskid surface.

3.4.2 Articulation lock. An articulation lock in accordance with SAE J276 shall be installed.

3.4.3 Noise level. Noise level determinations shall be made in accordance with MIL-STD-1474, sections 5.1.2.3.1 Vehicles and 5.1.2.5.1 Test environment. Noise level in the cab shall not exceed 85 Db(A) with the door shut. If the noise level exceeds 85 Db(A) with the cab door wide open, a warning plate shall be permanently mounted in a conspicuous location in the cab. The warning plate shall have "WARNING" at the top and shall instruct the operator to use hearing protection when operating the unit with the cab door open.

3.4.4 Backup alarm. A backup warning alarm audible above the noise of the vehicle shall be installed.

3.4.5 Sharp edges. All nonfunctional edges shall be rounded or broken so as to not present a safety hazard.

3.5 Engine. The engine shall be a standard commercial diesel engine. The demand on the engine under any specified operating condition shall not exceed the previous intermittent horsepower and speed ratings of the engine. The engine shall be capable of delivering sufficient power to efficiently perform all specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

tasks under any specified conditions. The engine shall operate on JP-8 conforming to MIL-T-83133 without detrimental effect on the engine. The engine shall be certified that it has successfully completed NATO AEP-5 test procedures (400 hours dynamometer test to assure JP-8 compatibility).

3.5.1 Power to weight ratio. The Gross Vehicle Weight to Net Flywheel Horsepower ratio shall not exceed 360 pounds per horsepower.

3.5.2 Accessory equipment. All of the following equipment shall be installed on the unit:

a. Two stage dry air filter with indicator.
b. Oil filter.
c. Fuel filter(s).
d. Hour meter.
e. Fuel tank capacity sufficient to provide 8 hours operation.
f. Batteries, maintenance free, with no filler caps or low maintenance batteries.

3.6 Power train. The power train shall be capable of withstanding all loads and speeds encountered during all normal operation.

3.6.1 Transmission. A full power shift transmission possessing a minimum of four forward and one reverse speed ranges shall be installed.

3.6.2 Axles, differentials, and final drives. The scraper shall be equipped with a locking or limited slip differential, full floating axles, and planetary final drives.

3.7 Systems and components.

3.7.1 Construction. Bow, elevator, ejector, front and rear frames, associated linkages, and hydraulic cylinders and rams shall be steel. These parts shall be constructed to maintain their structural integrity for a minimum of twelve years under normal conditions of operation, transport, and storage.

3.7.2 Bowl. The bowl shall be capable of power down, and shall provide for a depth cut of at least 9 inches. The cutting edge shall be either fixed or integral with the retractable floor, and equipped with replaceable, reversible wear plates conforming to SAE J737 and J738. A minimum of four cutting edge teeth shall be provided. The bowl shall be equipped with reversible and interchangeable side bits. The bowl shall be rated according to SAE J957 at not less than 11 cubic yards capacity. Rated payload shall be at least 26,000 pounds.

3.7.3 Sliding floor and ejector. The combination sliding floor/ejector shall be equipped with double acting cylinders and shall provide for complete dumping of the bowl.

3.7.4 Elevator. The elevator shall be hydraulically driven, and shall provide at least two speeds forward and one speed reverse. A minimum of 15 flights shall be provided. The elevator to cutting edge distance shall be adjustable by the operator without the use of tools.

3.7.5 Hydraulic system. A hydraulic system shall be used for transmitting engine power to the mechanisms that operate the elevator, bowl, ejector, and steering. The hydraulic system shall be completed with all operating accessories including pump(s), reservoir(s), valves, filters, hoses, piping, fittings, and pressure valve(s). Pump(s) shall have sufficient capacity to energize all hydraulically powered components. The hydraulic lines and hoses shall be capable of withstanding a bursting pressure of four times the working pressure. The system, including the hoses, shall be suitable for operating in temperatures from -40° to 250°F. Hoses shall be protected against chaffing through the use of supports and protective devices. Hydraulic system shall be protected against the entrance of foreign material. A wiper shall be provided on each cylinder near the seal which shall prevent abrasives from damaging the seal and entering the cylinder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

System shall be furnished with full flow filters having a micron rating not greater than 25. When an open type hydraulic system is furnished, reservoir(s) shall be vented through a separate vent or through the filler cap. In either case, the vent shall be equipped with a filter element to prevent the entrance of foreign material. Filter shall have a micron rating of not greater than 25. Reservoir(s) shall have sufficient capacity to allow for maximum displacement of all cylinders at the same time and to provide the necessary cooling capacity. Return line shall be terminated below the minimum fluid level of the reservoir. Suction lines shall be above the reservoir bottom. On open type hydraulic systems, the suction line shall be protected with a readily accessible strainer. Three separate control valves shall be provided to permit independent control of the elevator, ejector, and bowl. The elevator valve shall provide a minimum of two forward speeds and one reverse. The ejector valve shall provide control for eject, hold, and return. The bowl valve shall provide control for raise, power lower, and hold.

3.7.6 Hydraulic system priority. Hydraulic system shall be such that no reduction in usable steering ability occurs during reduction in engine speed or during the actuation of any other hydraulic controls.

3.7.7 Electrical system. The manufacturer's standard 12 or 24 volt electrical system shall be provided. The system shall be protected against overload by circuit breakers. Cables shall be bundled and wrapped or covered where possible. Cables and bundles shall be secured at least every three feet. Rubber grommets or boots shall be provided to protect cables from chafing where they pass through openings in the frame or other metal surface.

3.7.8. Braking system. Service brakes shall be installed on all four wheels. Braking performance shall meet the requirements of SAE J1152 standards for service, emergency, and parking brake systems.

3.7.9 Components. The following components shall be installed.

a. Rear push block.
b. Flood lights sufficient to permit safe and efficient operation of the scraper in the dark.
c. Type E-2 tires conforming to The Tire and Rim Association Inc Year Book recommendations.

3.7.10 Fastener Quality Assurance. All fasteners of Grade 5 or higher shall meet the quality assurance requirements of U.S. Air Force Drawing No. 8838302.

3.8 Dimensions. The following criteria shall be met:

a. Maximum depth of cut at least 9 inches.
b. Cut width at least 90 inches.
c. Maximum depth of spread at least 14 inches.
d. Minimum ground clearance at least 14 inches.

3.9 Performance.

3.9.1 Steering. The steering control shall be capable of 90 degrees right and left articulation.

3.9.2 Fuel capacity. The scraper shall have sufficient fuel capacity to operate continuously for 8 hours.

3.9.3 Speed. Loaded scraper shall be capable of 17 mph over a hard surface up a two percent grade.

3.9.4 Slopes.

a. The unloaded scraper shall be capable of 360 degrees of maneuverability on a thirty percent slope without loss of any hydraulic functions.
b. The scraper, when loaded to struck capacity, shall be capable of 360 degrees of maneuverability on a twenty percent slope without loss of stability.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:45Z
Check the source to verify that this is the current version before use.

3.9.5 Loading. Scraper shall be capable of self loading to heaped capacity in a period of one minute in undisturbed soil.

3.9.6 Draw bar pull. Scraper shall have a draw bar pull of not less than 25,000 pounds at 1.4 mph. Tire slippage shall not exceed eight percent on a brushed paved surface.

3.9.7 Ejector. Ejector mechanism shall stroke sufficiently to completely empty the bowl of loose material.

3.10 Operator compartment. The contractor's standard or optional ROPS cab which meets the performance requirements of SAE J1040 shall be provided.

3.10.1 Seat. Operators seat shall be equipped with a safety belt. Seat shall be sufficiently adjustable to allow comfortable operation for individuals ranging from 5'4" to 6'6" in height. Seat shall have hydraulic or air suspension. Upholstery shall be dark, extreme service material.

3.10.2 Controls. A three lever control group for elevator, ejector, and bowl shall be installed. The controls shall operate pilot valves, which in turn, actuate the main valves. All grouped controls shall have approximately equal throw distances, dead zones, and resistances to movement. All controls shall have a smooth feel and shall not bind, nor have more than 0.125 inch of slop at the grip position.

3.10.3 Instruments, indicators, and gages. The following instruments, indicators, or gages shall be installed:

a. Engine oil pressure gage or warning indicator.
b. Engine coolant temperature gage or warning indicator.
c. Transmission/torque converter oil temp gage or warning indicator.
d. Ammeter or voltmeter or warning indicator.
e. Engine hourmeter or service meter.
f. Air pressure gage (for units equipped with air brakes).
g. Low air pressure warning buzzer (for units equipped with air brakes).
h. Fuel level gage.
i. Hydraulic oil level indicator (gage or dipstick).
j. Air cleaner restriction indicator.

All instruments, gages, and indicators shall be identified with universal symbols in accordance with SAE J298.

3.10.4 Heater and defroster. A cab heater of sufficient capacity to raise the cab temperature from 0° to 40°F in one hour shall be provided. A front windshield and rear window defroster shall be provided and may be integral with the heater.

3.10.5 Cab lighting. All instruments, indicators, gages, and controls shall be sufficiently illuminated to permit safe operation of the scraper in the dark.

3.10.6 Windshield wipers. Front and rear windshield wipers sized and positioned to provide clear vision to operate the vehicle in rain or snow shall be installed.

3.11 Optional equipment. Each item described in this section shall be installed when specified in the ordering data (see 6.2).

3.11.1 Winterization. Winterization provisions for starting down to -20°F and operating down to -40°F without the use of external power shall be installed. These provisions shall include, but are not limited to the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

a. Measured shot type ether injection system.
b. Alcohol evaporator kit for air brake systems.
c. Cab heater of sufficient capacity to raise the cab temperature from -20°F to +25°F in one hour.
d. Engine coolant shall be protected to -40°F.
e. 170°F to 180°F engine thermostat shall be installed.
f. Boots, grommets, and plugs shall be used to close all holes in the cab.
g. Full engine enclosure.
h. Unit shall be capable of withstanding a 72 hour soak at -65°F without damage.

3.11.2 Fenders. Fenders which provide stone throw protection to the rest of the scraper shall be installed on the front wheels.

3.11.3 Bumper and pull hooks. A front bumper shall be provided. A device(s) such as tow hooks, loops, tow tube, etc for extracting a bogged down unit shall be installed. The device shall be capable of transferring at least a force equal to the GVW of the unit.

3.11.4 Transmission guard. A transmission guard shall be installed which shall protect the transmission from damage caused by running over large obstacles. The installed guard shall be capable of withstanding forces encountered when the scraper, travelling at 3 mph and loaded to capacity, runs up on an obstacle which causes the front of the scraper to be supported only by the transmission guard. Maintenance access ports or hatches and drainage holes for water and system fluids shall be provided as necessary.

3.11.5 Engine enclosure. An engine enclosure shall be provided on nonwinterized scrapers.

3.12 Lifting provisions. Lifting provisions shall be provided which provide necessary attachment points to allow for commercial shipping by boat.

3.13 Tie down provisions. Tie down provisions shall be provided which provide the necessary attachment points for all applicable forms of commercial shipping.

3.14 Data plates. The following plates shall be installed in a conspicuous location on the end item.

3.14.1 Identification plate. A corrosion resistant metal name plate shall be permanently and legibly marked in accordance with MIL-STD-130 with the following information:

a. Nomenclature.
b. Manufacturer's model number.
c. Manufacturer's serial number.
d. Special characteristics (HP, RPM, etc).
e. National stock number.
f. Manufacturer's name.
g. Date of manufacture.
h. Warranty expiration date.
i. Curb weight and CG location.

3.14.2 Lubrication plates. A corrosion resistant metal lubrication plate(s) shall illustrate the lubrication points on the vehicle and shall identify the type of lubricant and service interval for each point. An illustrated chart in the operators manual providing the same information may be provided in lieu of the metal plate.

3.14.3 Warning plates. Warning plates shall identify potentially hazardous conditions, locations, and actions. Plates shall be sized and located so as to be conspicuous to the person at risk.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

3.14.4 Instruction plates. All controls shall be labeled to identify the location and function of each control position. Instructions cast or stamped on or beside the controls are acceptable.

3.15 Treatment and painting. Scrapers shall be painted in accordance with standard commercial practice provided that the paint and primer is both lead and chromate free. Color shall be 24052 of FED-STD-595 unless otherwise specified in 6.2. The exterior finish shall show no evidence of uneven application, runs, sags, curtains, orange peel, lack of adhesion, or other defect. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted to match the chassis.

3.15.1 Markings. Markings (US Air Force and Registration Numbers) will be applied on dark green vehicles utilizing black paint color Number 37038. Place registration numbers on each side of vehicle and on rear if space permits. Size of letters and numerals to be used are as follows:

a. US AIR FORCE 1-1/2 INCHES
b. REGISTRATION NUMBER 1-1/2 INCHES
c. REG NUMBER SPACING EXAMPLE 77D1977

3.16 Workmanship. The workmanship shall be of the quality prevailing among manufacturers normally producing equipment of this type. Welded, bolted, and riveted construction shall be in accordance with the highest industry standards.

4. QUALITY ASSURANCE.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Notification of testing. The Government shall be, in accordance with contract provisions, notified of all testing to be performed in fulfillment of the requirements of this specification so that arrangements may be made to witness the testing. Addresses and specific requirements of notifications shall be provided in the solicitation, contract, or by the Government's contracting officer upon request.

4.2 Classification of tests. Testing shall be of two types, First Article Testing and Acceptance Testing. First article testing, when required (see 6.2), shall be performed on a unit prior to acceptance of any unit by the Government. The first article test sample may be a preproduction item or a production item which meets the requirements of this specification. The sample shall be identical to the items procured under the contract. First article testing shall consist of the requirements in section 4.4. Acceptance testing shall be performed on each unit prior to acceptance of that unit. Acceptance testing shall consist of the requirements in section 4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

4.3 Test conditions.

4.3.1 Apparatus. The apparatus used in conjunction with the testing specified herein shall be of laboratory precision and shall be calibrated at proper intervals to ensure accuracy.

4.3.2 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all test specified herein, the vehicle shall be closely observed for the following conditions, any of which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.
b. Spillage or leakage of any fluid, including fuel, coolant, lubrication or hydraulic fluid, under any operation condition except as allow herein.
c. Structural failure, evidence of impending failure, or permanent deformation of any component or joint.
d. Evidence of excessive wear.
e. Interference between vehicle components or between the vehicle and the ground (including all required obstacles).
f. Misalignment of any component.
g. Evidence of undesirable roadability characteristics, including instability in handling while cornering, braking, and traversing rough terrain as required.
h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.
i. Overheating of the engine, transmission, or any other vehicle component.

4.4 First article testing.

4.4.1 Examination. The first article shall be examined for conformance to section 3 in accordance with the checklist in Table 1.

TABLE 1

EXAMINATION

NO.	DESCRIPTION	PARA NO.
101	Conventional scraper design	3.1
102	All commercial standard features provided	3.1.1
103	Unit completely assembled or in transport configuration.	3.1.2
104	Fluid filling, draining and checking	3.2.1
	Fill ports conspicuous and convenient	
	engine oil | | engine coolant | |	
	trans fluid | | hydraulic system(s) | |	
	Ability to check fluid level as specified	
	engine oil | | engine coolant | |	
	trans fluid | | hydraulic system(s) | |	

Source: https://assist.dla.mil -- Downloaded: 202640-23T09:45Z
Check the source to verify that this is the current version before use.

TABLE 1 (Continued)

EXAMINATION

NO.	DESCRIPTION	PARA NO.
	Draining as specified engine oil | | engine coolant | | trans fluid | | hydraulic system(s) | | Sampling valves as specified engine oil | | hydraulic system(s) | | trans fluid | |	
105	Lube points as specified	3.2.2
106	Scraper edge as specified	3.2.6
107	Hydraulic test points as specified	3.2.7
108	General safety as specified	3.4
109	Grab rails, steps, platform, etc. as specified	3.4.1
110	Articulation lock as specified	3.4.2
111	Sharp edges treated as specified	3.4.5
112	Engine equipment (e. to be verified by test)	3.5.2
113	Bowl as specified	3.7.2
114	Floor and ejector as specified	3.7.3
115	Elevator design and construction as specified	3.7.4
116	Hydraulic system as specified	3.7.5
117	Electrical system as specified	3.7.7
118	Components as specified	3.7.9
119	Dimensions as specified	3.8
120	ROPS cab as specified	3.10
121	Seat as specified	3.10.1
122	Controls, indicators, instruments, and gages as specified as specified	3.10.2, 3.10.3, 3.10.5, 3.14.4
123	Windshield wipers as specified	3.10.6
	Optional equipment as specified (see 6.2)	
124	Winterization provisions a. | | b. | | c. N/A d. | | e. | | f. | | g. | | h. N/A	3.11.1
125	Fenders	3.11.2
126	Bumper and pull hooks	3.11.3
127	Transmission guard installed	3.11.4
128	Engine enclosure	3.11.5
129	Lifting provisions as specified	3.12
130	Tie down provisions as specified	3.13
131	Identification plate as specified	3.14.1
132	Lubrication plate(s) as specified	3.14.2
133	Warning plates as specified	3.14.3
134	Paint as specified	3.15
135	Marking as specified	3.15.1
136	Workmanship as specified	3.16

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

4.4.2 Preliminary operation. The scraper shall be operated for a period of not less than 30 minutes. All controls shall be operated sufficiently to determine ease of operation, but not less than 10 times each. The scraper shall travel in each speed range for at least one minute. All lights shall be turned on for the duration of the test. Service, emergency, and parking brakes shall be applied. Starter disconnect shall be demonstrated. All cab climate control functions shall be operated sufficiently to show that they are functional. Horn and backup warning signal shall be demonstrated. Down shift inhibitor shall be demonstrated in all but the top two gears.

4.4.3 Maintenance testing. After having performed the examination according to 4.4.1, an authorized government representative shall select between four and all of the tasks listed in 3.2.8. These tasks shall be performed sequentially by one person in the sum of the time allotted for each task. All tools, parts, and supplies shall be on hand at the start of the test. The list of tasks performed and the total time taken to perform the tasks shall be recorded. The use of any tools other than common hand tools shall be noted. Power tools shall not be used.

4.4.4 Noise level testing. Noise level determinations shall be made in accordance with MIL-STD-1474, sections 5.1.2.3.1 Vehicles and 5.1.2.5.1 Test environment. Noise level shall be as specified in 3.4.3. Previous test data may be acceptable if readily comparable to tests conducted as specified. Approval of previous test results shall be obtained prior to first article testing if applicable.

4.4.5 Mobility tests.

a. 90 degrees left and right articulation shall be demonstrated (per 3.9.1).
b. Scraper shall demonstrate the ability to travel at a speed of 17 mph on a two percent slope when loaded with 26,000 pounds payload. The test course shall consists of a hard smooth surface with an average slope of between two percent and four percent not less than 250 yards long. The grade of the course shall not be less than one percent over any 100 foot section of the course. The gross vehicle weight of the scraper shall be adjusted to simulate a two percent grade according to the following formula:

$$\text{test GVW} = \frac{2\%\ \text{GRADE} + 2\%\ \text{ROLL RESISTANCE}}{\text{AVE TEST GRADE} + 2\%\ \text{ROLL RESISTANCE}} (26{,}000\# + \text{CURB WEIGHT})$$

The average grade shall be determined to an accuracy of .05 percent. Weight shall be measured with an accuracy of 500 pounds. Vehicle velocity shall be measured with an accuracy of .5 mph. The scraper shall enter the measured course at not less than 17 mph and shall complete the course with an average velocity not less than that which it entered. Entrance velocity shall be measured using an instantaneous velocity device. Average velocity shall be determined using distance/time calculations or from an instantaneous velocity integral.

c. The scraper shall demonstrate 360 degrees of maneuverability on a thirty percent slope. No loss of stability shall occur. No wheel shall leave the ground. All hydraulics shall be operated at least four times spaced out equally over the test. No loss of any function shall occur. No reduction in oil or hydraulic fluid pressure shall occur.
d. The scraper, when loaded to 75 percent of rated capacity with a payload weighing not less than 19,000 pounds shall demonstrate 360 degrees of maneuverability on a twenty percent slope. No damage or loss of stability shall occur. No wheel shall leave the ground.

4.4.6 Operational testing. The scraper shall accumulate a minimum of twenty hours of scraper operation in addition to that required by other tests specified. The scraper shall self load to rated capacity on each pass and shall haul a distance of between 1,500 and 2,500 yards during each pass. Hauling shall take place over well prepared haul roads.

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:45Z
Check the source to verify that this is the current version before use.

4.4.7 Heater testing. After a twelve hour soak at a temperature not greater than 0°F, the engine shall be started without the use of external power. The engine shall start within three minutes. The cab heater shall be turned on. The temperature of the air four inches above the floor in an area not subject to direct heater output shall not be less than 40°F one hour after the engine was started. Where winterization is specified, the temperature of the cold soak shall not exceed -20°F and the heater shall raise the temperature, measured as described above, to not less than 25°F in one hour.

4.4.8 Lifting provisions testing. The static lifting load on each fitting shall be measured or calculated. A load of 1.5 times the static lifting load shall be applied to each fitting for a period of one minute. Each fitting shall be examined for failure and permanent deformation.

4.4.9 Tie down provisions testing. The scraper shall be tied down to a flat bed railroad car. The stationary flatbed containing scraper shall be bumped by two or more similarly sized cars traveling at not less than seven miles per hour. The scraper shall be examined for failure and permanent deformation.

4.4.10 Final inspection. The scraper shall again be operated after completing 4.4.2 through 4.4.9. All controls shall be operated sufficiently to determine ease of operation, but not less than 10 times each. The scraper shall travel in each speed range for at least one minute. All lights shall be turned on for the duration of the test. Service, emergency, and parking brakes shall be applied. All cab climate control functions shall be operated sufficiently to show that they are functional. The scraper shall be examined for signs of any conditions for failure listed in 4.3.4.

4.4.11 Test report. The following shall be incorporated as a part of a test report:

a. All failures, maintenance performed, and other events recorded shall be identified by accumulated operation time, miles, cycles, or position of the test procedure, as appropriate. Test conditions, at the time of failures or irregular operations identified, shall be recorded.
b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.
c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made, and the predicted effectiveness of those corrections.
d. Test activity and/or contractor comments on item features or requirements, that if modified, should improve the item.
e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.
f. The contractor shall furnish the following certifications. Each shall be an original signed by an authorized OEM representative and by an authorized contractor representative and shall clearly identify the type of item being certified, the limits to the certification, and the end item to which it will be installed (i.e. 11 yd scraper):

(1) Certification of scraper volume as specified in 3.7.2
(2) Certification of 26,000 pound capacity supported by commercial specification sheets.
(3) Certification that articulation lock meets SAE J276 as specified in 3.4.2.
(4) Certification of engine power rating for scraper duty cycle. Commercial specification sheet shall be included. Calculations showing conformance to 3.5.1 shall be included.
(5) Certification of transmission torque rating in excess of the maximum driving and breaking torques achievable. Certification of transmission speed rating in excess of operational speed.
(6) Certification that each scraper shall possess a differential as specified in 3.6.2.
(7) Certification that hydraulic lines and hoses have a bursting pressure at least four times the working pressure.
(8) Certification that ROPS cab has been tested and meets the requirements of SAE J1040.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

(9) Certification that transmission guard, if specified on any item in the contract (see 6.2), meets the requirements of 3.11.4.
(10) Certification that the coating system is as specified in 3.15 and 3.15.1.

4.5 Acceptance testing.

4.5.1 Inspection. Each scraper shall be inspected according to Table 1, for Items 103 and 121 through 136 inclusive.

4.5.2 Operation. The scraper shall be operated for a period of not less than 30 minutes. All controls shall be operated sufficiently to determine ease of operation, but not less than 10 times each. The scraper shall travel in each speed range for at least one minute. All lifts shall be turned on for the duration of the test. Service, emergency, and parking brakes shall be applied. Starter disconnect shall be demonstrated. All cab climate control functions shall be operated sufficiently to show that they are functional. Horn and backup warning signal shall be demonstrated. Down shift inhibitor shall be demonstrated in all but the top two gears.

4.5.3 Final examination. The scraper shall be examined for signs of any conditions for failure listed in 4.3.4.

4.6 Refurbishment. After successful completion of all testing and prior to delivery, each unit shall be refurbished to a like-new condition.

5. PREPARATION FOR DELIVERY.

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These scrapers are intended for military construction and landfill operations. They may be stored, transported, and operated in all environments. They may be subjected to extremes in temperatures and humidity and exposed to extremely corrosive coastal environments. Scrapers may be self loaded and loaded by auxiliary equipment in all ground conditions found in land clearing, earth moving, and landfill operations. The scrapers will scrape, transport, and dump all material found in land clearing, earth moving, and construction applications.

6.2 Ordering data.

a. Title, Number, and Date of this specification.
b. Name or number of any applicable amendments or supplements.
c. Winterization (see 3.11.1).
d. Fenders (see 3.11.2).
e. Bumper and towing device (see 3.11.3).
f. Transmission guard (see 3.11.4).
g. Engine enclosure. (see 3.11.5).
h. Color, if different than specified (see 3.15).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 Report. If a first article test report is required, see 4.4.11.

6.4 Subject term (key word) listing.

Bowl
Draw bar pull
Ejector
Hydraulic

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project number:
3805-F134

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-83807	90 APR 27

3. DOCUMENT TITLE

Scraper, Self Propelled, 11 Cubic Yards Capacity, Self Elevating

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME WR-ALC/MMEDT	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* Robins AFB, GA 31098-5609	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-01-23T04:45:14
Check the source to verify that this is the current version before use.